Chirag Vijay Patil

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Education

Virginia Tech, Blacksburg **Aug 2023 - May 2025**
Master of Science, Computer Engineering GPA: 3.96/4
Relevant Coursework: Data Analytics, Computer Vision, Large Scale Software Engineering, Advanced Machine Learning and Deep Learning, Application of Machine Learning, Database Management Systems

University of Mumbai, India **Aug 2018 - Jun 2022**
Bachelor of Engineering in Electronics and Telecommunications Engineering GPA: 8.36/10
Relevant Coursework: Big Data Analytics, Object Oriented Programming using Java, Digital System Design

Technical Skills

Programming Languages: **Python**, SQL, JavaScript, HTML/CSS, **C++**

Deep Learning: PyTorch, TensorFlow, JAX, scikit-learn, SciPy, OpenCV, Vision Transformers, Large-Scale Model Training

Databases Technologies & Frameworks: MySQL, PostgreSQL, **Django**, **ReactJS, NodeJS**, Flask, Apache Spark, FastAPI

Work Experience

Graduate Research Assistant (Chirag Patil - SANGHANI Center, Virginia Tech) **Blacksburg, VA, United States**
Li Lab of Applied Machine Learning in Genomics and Phenomics Sep 2024 - Present

- Leading the development of a fine-tuned **multimodal** Grounded DINO (GDINO) model, leveraging **vision transformers** for precise object detection in confocal root images to enhance the accurate identification of plant root cells, achieving a **20% improvement** in detection accuracy.
- Integrating the Segment Anything Model (microSAM) to refine segmentation accuracy, reducing segmentation errors by **30%** and enhancing object localization.
- Worked on a **hyperspectral datase**t with over **500,000 data points** to enable the effective use of the Segment Anything Model (SAM) for improved segmentation, resulting in a **25% increase** in segmentation precision.

Cyber Security Intern **Mumbai, India**
Tele-Networks Technologies Pvt. Ltd Jan 2022 - Feb 2022

- Improved network performance and security by **30%** through in-depth information gathering using DNS, SNMP, and SMTP, collaborating with cross-functional IT teams to enhance security measures.
- **Strengthened** cybersecurity protocols by mastering router configuration, network sniffing (Wireshark), and security tools (firewalls, IDS/IPS, SIEM, endpoint protection), reducing vulnerabilities by **40%.**

IT Intern **Mumbai, India**
Code Wind Jun 2021 - Jul 2021

- Collaborated with team to design and implement the authentication module, ensuring secure user access management.
- Developed role-based access control to restrict user permissions based on predefined roles, enhancing security.
- Designed database storage and UI interactions to efficiently manage user authentication and authorization.

Projects

Stock Portfolio Management System| Django, JWT, AWS, Docker, Git, Apache, Nginx, Redis, Celery **Aug 2024 – Dec 2024**

- Implemented OAuth2 using JWT for Google authentication and secured communication with HTTPS, improving security and preventing cross-site forgery.
- Designed a distributed task scheduling system using Celery and Redis, reducing server load by 40% and enabling real-time price tracking with user notifications via SMTP, thus improving user engagement
- Dockerized and orchestrated containers using Kubernetes to include seamless integration to host on AWS EC2

Brain Tumor Semantic Segmentation using Deep Learning | Python, PyTorch, OpenCV, scikit-learn **Jan 2024 – May 2024**

- Implemented **deep learning models**, including **U-Net** and **modified U-Net** architectures with **attention mechanisms**, for semantic segmentation of brain tumors from **multimodal MRI scans**.
- Pre-processed and normalized medical imaging data (MRI scans) from the BraTS dataset, involving tasks such as loading, normalizing, cropping, and splitting data into training and validation sets.
- Contributed to a comparative analysis of different methodologies, including traditional object detection (YOLOv9) and **transfer learning** approaches, for brain tumor detection and segmentation.

Soil Type Classification using Computer Vision | Python, Tensorflow, Open CV, scikit-learn **Sep 2023 – Dec 2023**

- Researched and implemented two distinct methodologies for soil type classification, one centered on **image processing** and **machine learning**, the other leveraging **Convolutional Neural Networks**; achieved an accuracy improvement of **15% using CNN-based approach**.
- Conducted extensive experiments and analysis, evaluating performance metrics and the impact of data augmentation to determine the most effective approach.